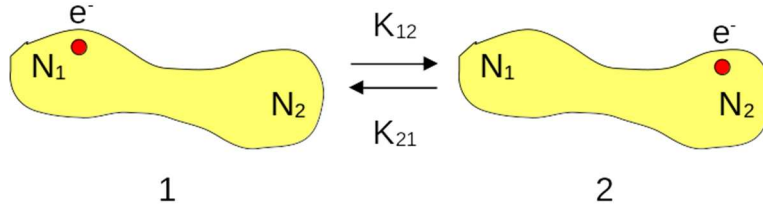


CW-ESR spectrum of a radical with two limiting configurations



Following Schneider and Freed [*Adv. Chem. Phys.* **1989**, 73, 387], from linear response theory the cw-ESR spectrum can be calculated as

$$G(\Delta\omega) \propto \Re \left\{ \langle\langle v | \hat{A}^\times{}^{-1} | w \rangle\rangle \right\} \quad (1)$$

with the super-operator

$$\hat{A}^\times = i\Delta\omega \mathbf{1} + i\hat{H}^\times + \hat{\Gamma} \quad (2)$$

where $\Delta\omega = \omega - \omega_0$, ω_0 is the Larmor frequency of the electron, $\hat{H}^\times$ is the spin super-Hamiltonian, and $\hat{\Gamma}$ an operator that influences spin relaxation due to the dynamics of the nuclei. In the Redfield limit, such an operator simply becomes a constant that accounts for line broadening as $R = 1/T_2$.

Also, in equation 1 are defined $|w \rangle\rangle \propto |v \rangle \otimes \rho_{eq} \rangle\rangle$, $|v \rangle\rangle = |\hat{S}_x \rangle \otimes \mathbf{1}_{\text{Nuclei}} \rangle\rangle$, where ρ_{eq} is the Boltzmann probability density, and $\mathbf{1}_{\text{Nuclei}}$ is the identity matrix with the same dimension of the direct product of the subspace of nuclei spin transitions.

In case of discrete jumps between two configurations, in the Redfield limit equation 1 reads

$$G(\Delta\omega) \propto \Re \left\{ \langle\langle \hat{S}_x \begin{bmatrix} 1 \\ 1 \end{bmatrix} \left| \begin{bmatrix} i\Delta\omega \mathbf{1} + i\hat{H}_1^\times - R_1 - K_{1,2} & K_{2,1} \\ K_{1,2} & i\Delta\omega \mathbf{1} + i\hat{H}_2^\times - R_2 - K_{2,1} \end{bmatrix}^{-1} \right| \hat{S}_x \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix} \rangle\rangle \right\} \quad (3)$$

where ρ_j is the Boltzmann weight of the j -th configuration, and $K_{j,j'}$ is the rate of jump from the j -th to the j' -th configurations.

Here we assume that in the j -th configuration the electron can interact with a single nucleus with moment of spin I_j . Also, we introduce the secular (high field) approximation. Therefore, the spin Hamiltonian reads

$$\hat{H}_j/\hbar = \omega_0 \hat{S}_z + \gamma_e A_j \hat{I}_{j,z} \hat{S}_z \quad (4)$$

where A_j is the hyperfine coupling constant between the electron and the j -th nucleus.

The simplest approach to calculate the spectrum from equation 3 is to span the super-operators over the direct product of spin transitions of the particles involved in the spin Hamiltonian:

$$|\sigma \gg = |\sigma^{(S)} \gg \otimes |\sigma^{(I_1)} \gg = |m_S, m'_S \gg \otimes |m_I, m'_I \gg \quad (5)$$

It is useful to remember that in the Liouville space

$$\ll \hat{O} |m, m' \gg = \langle m | \hat{O} | m' \rangle \quad (6)$$

$$\ll m_\mu, m'_\mu | \hat{O}_1 \cdot \hat{O}_2 | m_\nu, m'_\nu \gg = \langle m_\mu | \hat{O}_1 | m_\nu \rangle \langle m'_\mu | \hat{O}_2 | m'_\nu \rangle \quad (7)$$

$$\hat{O}^\times = \hat{O} \cdot \mathbf{1} - \mathbf{1} \cdot \hat{O}^\dagger \quad (8)$$

With the rules in equations 6-8 valid for generic linear operators $\hat{O}$, it is possible to transform equation 3 in matrix form $G(\Delta\omega) \propto \Re\{v^\dagger A^{-1} w\}$ with

$$A_{\mu,\nu} = \delta_{\mu,\nu} \begin{bmatrix} i\Delta\omega + ih_{1,\mu}^\times - R_1 - K_{1,2} & K_{2,1} \\ K_{1,2} & i\Delta\omega + ih_{2,\mu}^\times - R_2 - K_{2,1} \end{bmatrix} \quad (9)$$

with

$$h_{j,\mu}^\times = \omega_0(m_{S,\mu} - m'_{S,\mu}) + \gamma_e A_j (m_{I_j,\mu}, m_{S,\mu} - m'_{I_j,\mu}, m'_{S,\mu}) \quad (10)$$

The starting vectors can be recast in a more symmetric way: $|v \gg, |w \gg \rightarrow |\tilde{v} \gg, |\tilde{w} \gg$ with

$$|\tilde{v} \gg = \frac{1}{\sqrt{[I_1][I_2]}} \begin{bmatrix} \frac{1}{\rho_1^2} \\ \frac{1}{\rho_2^2} \end{bmatrix} \hat{S}_x \quad (11)$$

with $[I] = 2I + 1$. The starting vector is easily spanned over the basis set functions

$$\tilde{v}_{\mu,j} = \langle\langle \sigma_\mu | \tilde{v} \rangle\rangle = \frac{1}{2} \frac{1}{\sqrt{[I_1][I_2]}} \begin{bmatrix} \delta_{m_{I_1,\mu}, m'_{I_1,\mu}} \rho_1^{\frac{1}{2}} \\ \delta_{m_{I_2,\mu}, m'_{I_2,\mu}} \rho_2^{\frac{1}{2}} \end{bmatrix} \left(c_{m'_{S,\mu}}^+ \delta_{m_{S,\mu}-m'_{S,\mu},1} + c_{m'_{S,\mu}}^- \delta_{m_{S,\mu}-m'_{S,\mu},-1} \right) \quad (12)$$

$$\text{where } c_m^\pm = \sqrt{\frac{3}{4} - m(m \pm 1)}.$$

The matrix form of equation 3, therefore reads

$$G(\Delta\omega) \propto \Re \left[\begin{bmatrix} \tilde{p}_1^{(1)} \\ \vdots \\ \tilde{p}_{N_s}^{(1)} \\ \tilde{p}_1^{(2)} \\ \vdots \\ \tilde{p}_{N_s}^{(2)} \end{bmatrix} \begin{bmatrix} i\Delta\omega + ih_{1,1}^\times - R_1 - K_{1,2} & & & & K_{2,1} & & \\ & \ddots & & & & \ddots & \\ & & i\Delta\omega + ih_{1,N_s}^\times - R_1 - K_{1,2} & & & & K_{2,1} \\ K_{1,2} & & & i\Delta\omega + ih_{2,1}^\times - R_2 - K_{2,1} & & & \\ & \ddots & & & \ddots & & \\ & & K_{1,2} & & & i\Delta\omega + ih_{2,N_s}^\times - R_2 - K_{2,1} \end{bmatrix}^{-1} \begin{bmatrix} \tilde{p}_1^{(1)} \\ \vdots \\ \tilde{p}_{N_s}^{(1)} \\ \tilde{p}_1^{(2)} \\ \vdots \\ \tilde{p}_{N_s}^{(2)} \end{bmatrix} \right] \quad (13)$$

The matrix $\mathbf{A}$ is complex, symmetric under the secular approximation. It has dimensions $n_A = 4 \times [I_1]^2 + 4 \times [I_2]^2$. In case of $I_1 = I_2 = 1$, the matrix dimensions are $n_A = 72$. While the computational effort for the vector-matrix-vector product in this case is low, and it is embarrassingly parallel in the $\Delta\omega$ axis, it is more convenient to diagonalize $\mathbf{A}$, especially to treat cases with more nuclei and/or more configurations.

To this purpose, it should be noted that $\mathbf{A} = i\Delta\omega \mathbf{1} + \mathbf{A}'$. The two matrices $i\Delta\omega \mathbf{1}$ and $\mathbf{A}' = \mathbf{A} - i\Delta\omega \mathbf{1}$ commute. Therefore, they share the same set of eigenvectors, $\mathbf{U}$, and the eigenvalues of $\mathbf{A}$, λ_n , are simply given by $\lambda_n(\Delta\omega) = i\Delta\omega + \lambda'_n$, with λ'_n the n -th eigenvalue of $\mathbf{A}'$. If $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_{n_A})$ is the matrix of the eigenvalues of $\mathbf{A}$, one can rewrite the spectrum as

$$G(\Delta\omega) \propto \Re\{\tilde{\mathbf{v}}^\dagger \mathbf{A}^{-1} \tilde{\mathbf{v}}\} = \Re\{\tilde{\mathbf{v}}^\dagger (\mathbf{U} \mathbf{\Lambda} \mathbf{U}^\dagger)^{-1} \tilde{\mathbf{v}}\} = \Re\{\tilde{\mathbf{v}}^\dagger \mathbf{U}^\dagger \mathbf{\Lambda}^{-1} \mathbf{U} \tilde{\mathbf{v}}\} = \Re\{\tilde{\mathbf{V}}^\dagger \mathbf{\Lambda}^{-1} \tilde{\mathbf{V}}\} \quad (14)$$

where it was used the fact that for a unitary transformation, $\mathbf{U}^{-1} = \mathbf{U}^\dagger$.

By making explicit the products in the rightmost equality of equation 14, one gets

$$G(\Delta\omega) \propto \Re e \left\{ \sum_{n=1}^{n_A} \frac{|V_n|^2}{\lambda'_n + i\Delta\omega} \right\} \quad (15)$$

Finally, the line shape, $I(\Delta\omega)$, is easily obtained as the derivative

$$I(\Delta\omega) = -\frac{dG(\Delta\omega)}{d\omega} \propto \Re e \left\{ \sum_{n=1}^{n_A} i \frac{|V_n|^2}{(\lambda'_n + i\Delta\omega)^2} \right\} \quad (16)$$